$y = m_1x_1 + m_2x_2 + b$

$\hat{y} = mx + b$, $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $m_1 = -1$, $m_2 = 2$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$,

$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

x	y	$\hat{y}$	$\hat{y} - y$
$\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$	$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 6 \\ 17 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Getting $\hat{y}$ (yhat)

$\hat{y} = mx + b$

$\hat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} (-1 \times 1) + (2 \times 3) \\ (-1 \times 4) + (2 \times 10) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Calculating the ~~loss~~ MSE

$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$, $n = 2$

$MSE = \frac{1}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix}^2 \rightarrow \frac{1}{2} \left(\begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 11 \end{bmatrix} \right)$

$MSE = \frac{1}{2} \left[(1 \times 1) + (11 \times 11) \right]$

$MSE = \frac{1}{2} \times 122$

$MSE = 61$

Calculating the loss function(s)

Let MSE be J

$\frac{dJ}{dm_1} = ?, \frac{dJ}{dm_2} = ?, \frac{dJ}{db} = ?$

From

$$J = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}-y)^2$$

But

$$\hat{y} = m_1x_1 + m_2x_2 + b$$

So

$$J = \frac{1}{n}\sum_{i=1}^{n}((m_1x_1+m_2x_2+b)-y)^2$$

let $m_1x_1+m_2x+b-y$ be e

$$J = \frac{1}{n}\sum_{i=1}^{n} e^2$$

Applying chain rule

$$\frac{dJ}{dm_1} = \frac{dJ}{de}\cdot\frac{de}{dm_1}$$

$$\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e$$

$e = m_1x_1 + m_2x_2 + b - y$

$\frac{de}{dm_1} = x_1 + 0 + 0 - 0$

m_2, x_2, b, y behave as constants and go upon differentiation (become 0)

$$\frac{de}{dm_1} = x_1$$

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} e\cdot x_1$$

Since $e = (m_1x_1+m_2x+b-y)$, $\hat{y}$ and y

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n}(\hat{y}-y)x_1 \quad \text{— (1)}$$

$\frac{dJ}{dm_2} = ?$

$$\boxed{\frac{dJ}{dm_2} = \frac{dJ}{de} \cdot \frac{de}{dm_2}}$$ → Applying chain rule

$$J = \frac{1}{n}\sum_{i=1}^{n} (\hat{y} - y)^2$$

But $\hat{y} = m_1 x_1 + m_2 x_2 \overset{+b}{\cancel{y}}$

let $e = \hat{y} - y$

$e = m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}} - y$

$$J = \frac{1}{n}\sum_{i=1}^{n} e^2$$

$e = m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}} - y$

$$\boxed{\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e}$$

$\frac{de}{dm_2} = 0 + x_2 + 0 - 0$

m_1, x_1, b, y are constants and become 0 upon differentiation

$$\boxed{\frac{de}{dm_2} = x_2}$$

$$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} e \cdot x_2$$

$$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} \underbrace{(m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}} - y)}_{\hat{y}} x_2$$

$$\boxed{\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y) x_2} \quad \text{——— (11)}$$

$\frac{dJ}{db} = ?$

$J = \frac{1}{n}\sum_{i=1}^{n} e^2$, where $e = m_1x_1 + m_2x_2 + b - y$

$e = \hat{y} - y$

$\frac{dJ}{db} = \frac{dJ}{de} \cdot \frac{de}{db}$ → Applying chain rule

$\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e$

$\frac{de}{db} = 0 + 0 + 1 - 0$

$\frac{de}{db} = 1$

$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} e \cdot 1$ where $e = \underbrace{m_1x_1 + m_2x_2 + b}_{\hat{y}} - y$

$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)$ — (iii)

Therefore

$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)x_1$, $\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)x_2$, $\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)$

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $m_1 = -1, m_2 = 2$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$,

$x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$

$n = 2$

$$m_1 new = m_1 old - \alpha \frac{dJ}{dm_1}$$

$$m_1 new = m_1 old - \alpha \left[\frac{+2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_1 \right]$$

$$m_1 new = -1 - \alpha \left[\frac{+2}{2} \left[\begin{matrix} 1 \\ 11 \end{matrix} \right] \cdot \left[\begin{matrix} 1 \\ 4 \end{matrix} \right] \right]$$

$$m_1 new = -1 - \alpha \left[+1 \, (1 \times 1) + (11 \times 4) \right]$$

$$m_1 new = -1 - \alpha \left[+45 \right]$$

$$\boxed{m_1 new = -1 - 45\alpha} \quad \text{—(i)}$$

$$m_2 new = m_2 old - \alpha \left[\frac{2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_2 \right]$$

$$m_2 new = 2 - \alpha \left[\frac{2}{2} \left[\begin{matrix} 1 \\ 11 \end{matrix} \right] \left[\begin{matrix} 3 \\ 10 \end{matrix} \right] \right]$$

$$m_2 new = 2 - \alpha \left[(1 \times 3) + (11 \times 10) \right]$$

$$m_2 new = 2 - \alpha \left[113 \right]$$

$$\boxed{m_2 new = 2 - 113\alpha} \quad \text{—(ii)}$$

$$b_{new} = b_{old} - \alpha \frac{dJ}{db} \qquad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$b_{new} = b_{old} - \alpha \left[\frac{2}{n} \sum_{i=1}^{n} (\hat{y} - y) \right]$$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \alpha \left[\frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \right]$$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 1 \\ 11 \end{bmatrix} \quad \text{—— (iii)}$$

$m_{1new} = -1.45\alpha$, $m_{2new} = 2 - 113\alpha$, $b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Let $\alpha = 0.01$

$m_{1new} = -1.45\alpha$

$m_{1new} = -1.45(0.01)$

$m_{1new} = -1.45$

$m_{2new} = 2 - 113\alpha$

$m_{2new} = 2 - 113(0.01)$

$m_{2new} = 0.87$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ 0.11 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

MSE when $m_1 new = -1.45$, $m_2 new = 0.87$, $b new = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$

$\hat{y} = mx + b$ $\quad m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (-1.45 \times 1) + (0.87 \times 3) \\ (-1.45 \times 4) + (0.87 \times 10) \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$\hat{y} - y = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2 \qquad n = 2$$

$$MSE = \frac{1}{2} \left[\begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}^2 \right]$$

$$MSE = \frac{1}{2} \left((-2.85)^2 + (-2.21)^2 \right)$$

$$MSE = \frac{1}{2} \times 13.01$$

$$MSE = 6.51$$

∴ Therefore

$m_1 new = -1.45$

$m_2 new = 0.87$

$b new = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$

$MSE update = 6.51$

Daniel GAKUMBA NGWALE

$y = m_1 x_1 + m_2 x_2 + b$

$\hat{y} = mx + b$

Iteration 2

$$m = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} \quad b = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix}$$

$$Xm = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} = \begin{pmatrix} -1.45 + 3\times(0.87) \\ 4\times(0.87)\ 4\times(-1.45) + 10\times(0.87) \end{pmatrix} = \begin{pmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{pmatrix}$$

$$Xm = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix}$$

$$\hat{y} = Xm + b = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix} + \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix}$$

$$E = \hat{y} - y = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$E = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

~~$MSE = 1/n (E^2) = 1/n ($~~

$$MSE = 1/n (E_1^2 + E_2^2) = 1/2 ((-2.85)^2 + (-2.21)^2)$$

$$MSE = \frac{8.1225 + 4.8841}{2} = \frac{13.0066}{2} = 6.5033$$

$$d\,MSE/dm = 2/n (X^T E) = (2/2)\left[\begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}\right] = \begin{bmatrix} -2.85 + (4)(-2.21) \\ 3(-2.85) + (10)(-2.21) \end{bmatrix}$$

$$d\,MSE/dm = \begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix}$$

$$\frac{dMSE}{db} = \frac{2}{n} E = \frac{2}{2} \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$\frac{dMSE}{db} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$m_{new} = m_{old} - \alpha \frac{dMSE}{dm} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - 0.01 \begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix}$$

$$m_{new} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - \begin{pmatrix} -0.1169 \\ -0.3065 \end{pmatrix} = \begin{pmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{pmatrix}$$

$$m_{new} = \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix}$$

$$b_{new} = b_{old} - \alpha \frac{dMSE}{db} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} - 0.01 \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

$$b_{new} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} - \begin{pmatrix} -0.0285 \\ -0.0221 \end{pmatrix} = \begin{pmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{pmatrix}$$

$$b_{new} = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix}$$

Michael Odhiambo - Iteration 3

$$m = \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix} \quad b = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix} \quad x = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \quad y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

Step 1. $x \cdot m$

$$xm = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix} = \begin{pmatrix} 1(-1.3331) + 3(1.1765) \\ 4(-1.3331) + 10(1.1765) \end{pmatrix} = \begin{pmatrix} -1.3331 + 3.5295 \\ -5.3326 + 11.7650 \end{pmatrix} = \begin{pmatrix} 2.1964 \\ 6.4326 \end{pmatrix}$$

Step 2. $\hat{y} = x \cdot m + b$

$$\hat{y} = \begin{pmatrix} 2.1964 \\ 6.4326 \end{pmatrix} + \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix} = \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix}$$

Step 3 error -> $E = \hat{y} - y$

$$E = \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

Step 4 MSE

$$= \frac{1}{2} \begin{pmatrix} -1.7851 & 1.3447 \end{pmatrix} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$= \frac{1}{2} (3.1866 + 1.18082)$$

$$= \frac{1}{2} (4.9948) = 2.4974$$

Step 5 Gradient for m

$$(2/2) \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$= 1 \begin{pmatrix} 1(-1.7851) + 4(1.3447) \\ 3(-1.7851) + 10(1.3447) \end{pmatrix}$$

$$= \begin{pmatrix} 2.5937 \\ 8.0917 \end{pmatrix}$$

Step 6 Gradient for b

$$= (2/2) \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$= \begin{pmatrix} 1.7851 \\ 1.3447 \end{pmatrix}$$

Step 7 update m $m_{new} = m_{old} - \alpha \cdot \frac{dJ}{dm}$

$$m_{new} = \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix} - 0.01 \begin{pmatrix} 3.5937 \\ 8.0917 \end{pmatrix}$$

$$= \begin{pmatrix} 1.3331 \\ 1.2765 \end{pmatrix} - \begin{pmatrix} 0.0359 \\ 0.0809 \end{pmatrix}$$

$$= \begin{pmatrix} 1.3690 \\ 1.0956 \end{pmatrix}$$

Step 8 update b $b_{new} = b_{old} - \alpha \frac{dJ}{db}$

$$b_{new} = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix} - 0.01 \begin{pmatrix} 1.7851 \\ 1.3667 \end{pmatrix}$$

$$= \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix} - \begin{pmatrix} 0.0179 \\ 0.0134 \end{pmatrix}$$

$$= \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

Step 9 MSE after the update

$$\hat{y} = x \cdot m_{new} + b_{new} = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3690 \\ 1.0956 \end{pmatrix} + \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

$$= \begin{matrix} 1(-1.3690) + 3(1.0956) \\ 4(-1.3690) + 10(1.0956) \end{matrix} + \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

$$= \begin{matrix} 1.9178 \\ 5.4800 \end{matrix} + \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix} = \begin{pmatrix} 2.9542 \\ 6.3787 \end{pmatrix}$$

$$E = \hat{y} - y$$

$$= \begin{pmatrix} 2.9542 \\ 6.3787 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix}$$

$$MSE = 1/2 \begin{pmatrix} -2.0458 & 0.3787 \end{pmatrix} \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix}$$

$$= \frac{4.1853 + 0.1434}{2}$$

$$MSE = 2.1644$$

Therefore

m_1 new $= -1.3690$

m_2 new $= 1.0956$

$b_{new} = \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$

$MSE_{update} = 2.1644$